



Divit Mittal

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● EDUCATION & TRAINING

10/09/2022 - Current - JAIPUR, RAJASTHAN, INDIA

BACHELOR OF TECHNOLOGY IN DATA SCIENCE & ENGINEERING - MANIPAL UNIVERSITY

National classification: NCC5 | **Website:** <https://jaipur.manipal.edu/>

01/09/2021 - 01/09/2022 - DELFT, NETHERLANDS

BACHELOR OF SCIENCE IN COMPUTER SCIENCE & ENGINEERING - DELFT UNIVERSITY OF TECHNOLOGY

Final grade: 8.0 | **Type of credits:** ECTS | **Number of credits:** 20 | **Website:** tudelft.nl

31/03/2018 - 12/07/2020 - NEW DELHI, INDIA

ALL INDIA SENIOR SCHOOL CERTIFICATE EXAMINATION - CENTRAL BOARD OF SECONDARY EDUCATION (CBSE)

1. English Language - C1 level (CEFR) : A1 Grade
2. Mathematics : A1 Grade
3. Physics : A1 Grade
4. Chemistry : A1 Grade
5. Computer Science - C++ : A1 Grade
6. German Language - B2 level (CEFR) : A1 Grade
7. Work Experience - Robotics and Mechatronics : A1 Grade
8. Health & Physical Education : A1 Grade
9. General Studies - Sociology, Civics, and Cultural Studies : A1 Grade

Diploma: <https://europa.eu/!yp79jh> | **Transcript :** <https://europa.eu/!BB98Ww> | **Character Certificate:** <https://europa.eu/!HD84fw>

Final grade: A1 : ~96% | **National classification:** NSQF level 4 - (NCC 2) | **Address:** Delhi Public School, R.K. Puram

31/03/2016 - 28/05/2018 - NEW DELHI, INDIA

ALL INDIA SECONDARY SCHOOL EXAMINATION - CENTRAL BOARD OF SECONDARY EDUCATION (CBSE)

1. English Language - B2 level (CEFR) : A1 Grade
2. German Language - B1 level (CEFR) : A2 Grade
3. Mathematics : A1 Grade
4. Science - Physics, Chemistry, and Biology : A1 Grade
5. Social Science - History, Geography, Civics, and Economics : A1 Grade
6. Foundation of Information Technology : A1 Grade

Diploma: <https://europa.eu/!Mv98jr> | **Transcript:** <https://europa.eu/!uG83GN>

Final grade: A1 : ~97% | **National classification:** NSQF level 2 - (NCWP 2) | **Address:** Bhatnagar International School,

HONOURS AND AWARDS

30/09/2024 BobbleAI Keyboard, Gurgaon, Haryana
BobbleAI Datathon Winner Processed raw product data, i.e. [BobbleAI Keyboard](#) (7Cr+ users) event records in the last 5 days before July 1, 2024,i.e., about 8 million rows. Extracted & transformed JSON, to preprocess, ML-filter and store as structured **Apache Parquet** on **AWS S3** via serverless **AWS Lambda** functions & **PySpark** on **AWS Glue** .

31/10/2023 Alliance University, Anekal, Karnataka

- Leverage the high-fidelity **CARLA** simulator to create diverse driving scenarios for an **autonomous vehicle driving system**with RGBA & LiDAR sensors. Utilizing deep learning techniques, i.e. **CNNs** for **semantic segmentation**to improve the vehicle's understanding of its surroundings, & manouvre.
- A **Driver Drowsiness Detection** system that uses **Siamese Neural Networks** and computer vision techniques, lik**aarcascade** wave filters to monitor a driver's facial features in real-time and detect signs of fatigue. By analyzing eye states, & mouth movements, the system identifies potential drowsiness and enhances road safety.

Links <https://github.com/DivitMittal/CARLA-Autonomous-Driving>

30/01/2020 The Science Olympiad Foundation, New Delhi
International Rank 7 at the International Mathematics Olympiad

30/12/2019 The British Council Division, British High Commission, New Delhi
International Rank 49 at the International English Olympiad (IEO)

31/01/2019 Vijnana Bharati (VIBHA), with Vigyan Prasara, under the Department of Science and Technology & NCERT
India National Level Runner-up at the Vidyarthi Vigyan Manthan

LANGUAGE SKILLS

Mother tongue(s): **HINDI**

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
GERMAN/DEUTSCH (GOETHE-ZERTIFIKAT B2)	B2	B2	B2	B2	B2
ENGLISH (IELTS)	C2	C2	C1	C2	C1

SKILLS

Machine Learning & Data Science (Mathematics)
Deep Learning: Keras & PyTorch | Machine Learning: scikit-learn, tensorflow | LLM: Transformers, RAG, Encoders, & Decoders | Big Data: Hadoop, Spark, Hive, Kafka | Computer Vision: OpenCV, ITK

DevOps & Linux SysAdmin
NixOS Expert | IaC: nix, ansible & terraform | VCS: Git & Hg | Cloud Experience: AWS, GCP & Firebase | Shells: bash, zsh, fish, pwsh | Virtualization & Containerization: Proxmox, KVM, Docker & Podman | CI/CD: GitHub Actions & Jenkins

● PROJECTS

04/2024 - CURRENT

OS-nixCfg: A Personal Infrastructure as Code configuration **Nix**, nix-darwin, **NixOS**, home-manager, etc. configurations to attain my **deterministically reproducible** OS, written in the **functional** "Nix" expression language. Maintains:

- Darwin-kernel systems, i.e. macOS
- Android systems
- Desktop Linux-kernel systems, e.g. NixOS, Arch Linux, RHEL
- WSL2 on Windows

Links <https://github.com/DivitMittal/OS-nixCfg>

02/2024

Programming Languages Documentation Conversational AI @ IEEE MUJ RAG -enhanced **LLaMa-like transformer** neural network based LLM for assistance on programming languages/frameworks' documentations, with a **streamlit** front-end. Everything was implemented from scratch:

- Pre-trained **LLaMa-like transformer** with capability from **tokenization** to **embedding** (positional & word2vector) to output probabilities.
- **RAG** was implemented via **Qdrant** vector database on a **DigitalOcean Docker** container.

Links <https://github.com/DivitMittal/DocAssist-LLM>

10/2023

Generation of "Hazard Map" for safely navigating a lunar lander Is implemented by utilising the sensory data from the **ISRO**'s Chandrayaan-2 TMC2 & OHRC payloads for **SIH** (Smart India Hackathon). The project entails wrangling multiple data in form of multispectral high-res. imagery & stereo-triplet panchromatic data of lunar terrain. These multiple streams of data are fused together and analysed respectively for ground truth via **SRGANs** & **DCNNs**.

12/2021

Researched & designed wearable technology for the visually-impaired @ TU Delft An innovation which helps the blind to manoeuvre with agility and confidence, by ascertaining their environment, utilizing the help of ultrasonic waves and notifying haptically. The preeminent peculiarity of this innovation is, that it's affordable for everyone; built using **Arduino**, the total cost is less than €20.

11/2020

Developed an employee management cross-platform mobile application @ SASWA Benefiting from the advent of multi-platform deployment from a single source code, i.e. via **Flutter** & **Dart** aided me create an employee management iOS and Android application. Incorporating **Firebase** auth and database and thus, ameliorating the company's day-to-day inconveniences.

Links <https://github.com/DivitMittal/employee-management>